

Line Graphs Are Here! Unlock a Simple Solution for Data Sparsity and Class Imbalance in Recommender Systems

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Abstract

The persistent challenges of data sparsity and class imbalance have long limited the development of recommender systems. Fortunately, line graph theory offers a novel perspective to overcome these issues. By transforming the user-item interaction bipartite graph into a line graph, the problems of data sparsity and class imbalance are elegantly reformulated as those of insufficient labeled nodes and imbalanced label distribution in the line graph domain. This reformulation allows us to directly apply mature techniques from node classification and imbalanced graph learning to address these core challenges. Inspired by this insight, we propose a Line Graph Data Augmentation (LGDA) strategy, which features two distinct characteristics. Firstly, it is a plug-and-play module that resolves data sparsity and imbalance without modifying the underlying recommendation framework. Secondly, it employs a targeted augmentation and confidence filtering mechanism to generate high-quality, balanced augmented data. Extensive experiments on four real-world datasets validate that LGDA effectively alleviates data sparsity and class imbalance, leading to significant improvements in both recommendation performance and system robustness.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Recommender Systems; Line Graph; Data Augmentation

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Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18346766>.

1 Introduction

Recommender systems are fundamentally constrained by two intertwined challenges: data sparsity and class imbalance. Data sparsity describes the prevalent scenario where users interact with only a small subset of items, or items receive very few user interactions [4]. Class imbalance arises when the distribution of user-item interactions is highly skewed across categories [15]. Effectively addressing both issues is critical to achieving robust and high-performing recommender systems.

The detrimental impact of data sparsity is quantitatively evident. As shown in Figure 1a, as data sparsity increases, the RMSE of both FM and DNN models shows a clear upward trend. Although DNNs outperform FM in handling data sparsity due to their complex architectures and powerful learning capabilities, both approaches collapse when sparsity reaches 90%. This phenomenon underscores that without sufficient data, models struggle to learn user preferences and make accurate recommendations. Therefore, data sparsity is a key factor that prevents recommender systems from providing high-quality recommendations. Current mainstream approaches to mitigating data sparsity can be broadly categorized into two types: data augmentation and leveraging side information. Data augmentation [24] centers on generating simulated interactions derived from existing data. In contrast, leveraging side information

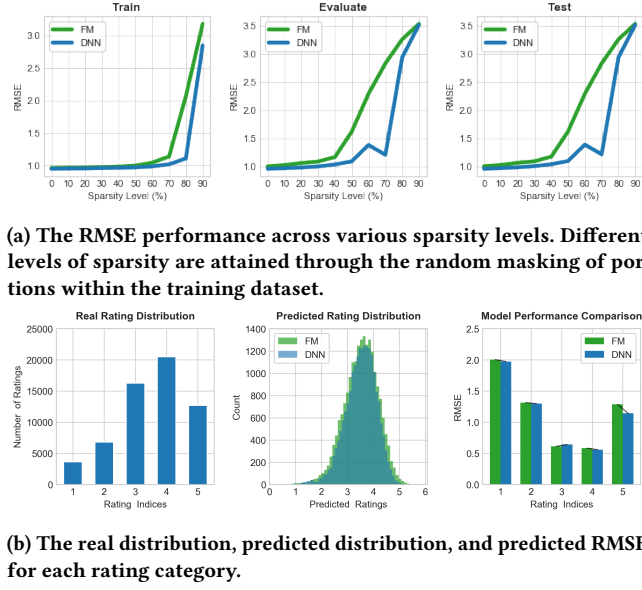


Figure 1: The empirical analysis of data sparsity and class imbalance in the MovieLens 100K dataset. FM and DNN are used as baselines. Lower RMSE means better results.

[6] emphasizes integrating external data sources, such as social networks or knowledge graphs. However, both approaches have inherent limitations: data augmentation may introduce noise if improperly applied, and side information is often context-specific, which limits its broader applicability.

Class imbalance, often manifested as a long-tail distribution (Figure 1b, left), presents a distinct yet equally critical challenge. It biases models toward popular (head) items, causing over-prediction for the head and under-prediction for the tail (Figure 1b, middle). Consequently, prediction errors are substantially higher for tail ratings (Figure 1b, right). In practical applications, however, misclassifying minority but high-value samples may lead to significant economic losses for platforms. For instance, consider an e-commerce platform with 1000 products to recommend, including 50 high-profit featured items. The core goal of the recommendation system is to accurately identify and recommend these 50 items. If the model predicts all products as non-featured, it can easily achieve 95% accuracy. Yet this would clearly defeat the purpose, as the model would fail to identify these 50 critical items, leaving users unable to access them. Current solutions to the long-tail recommendation problem are primarily categorized into data-level and algorithm-level methods. Data-level methods [28] balance class distribution via techniques like oversampling and undersampling. Algorithm-level methods [14] enhance focus on tail classes by dynamically adjusting loss functions or incorporating prior probabilities. In addition, some research attempts to adopt hybrid methods [3] to integrate the advantages of both data- and algorithm-level strategies.

The aforementioned challenges highlight the critical need to jointly address data sparsity and class imbalance in recommender

systems. Thankfully, line graph theory provides a promising pathway toward this goal [26]. In a line graph, each edge of an original graph becomes a node, and two such nodes are connected if their corresponding edges share a common endpoint [1]. In the recommendation context, user-item interactions naturally form a weighted bipartite graph [8]. Transforming this interaction graph into its line graph yields an insightful reformulation: each interaction (edge) becomes a node, and its rating becomes the node label. Consequently, the problem of data sparsity translates into a scarcity of labeled nodes, while class imbalance manifests as a skewed label distribution in the line graph domain. This reformulation unlocks a direct and powerful solution: we can now apply well-established node classification techniques to infer labels for unlabeled nodes (addressing sparsity), and employ imbalanced graph learning methods to rebalance the label distribution (addressing imbalance). Figure 2 provides an intuitive illustration of this conceptual transformation. To our knowledge, no relevant studies have yet attempted to apply this method in practice.

Building upon the above reformulation, we propose the Line Graph-based Data Augmentation (LGDA) strategy to jointly address data sparsity and class imbalance in recommender systems. The LGDA pipeline is carried out as follows. First, to control the computational cost, LGDA adopts a targeted augmentation mechanism that focuses on inactive users and unpopular items to identify augmentation candidates. The augmented interactions are then modeled as a weighted bipartite graph. Subsequently, for each target node pair to be augmented, its corresponding k -hop enclosing subgraph is extracted from this bipartite graph and further converted into a line graph. To mitigate class imbalance in the transformed line graph, synthetic minority-class nodes and their associated edges are generated via embedding interpolation, thus rectifying the skewed label distribution. The balanced line graph is then used to train an unbiased GNN classifier, which serves as the foundation for subsequent interaction prediction. Finally, a confidence-based filtering strategy is applied to refine the generated augmented data, ensuring its high quality and balanced distribution. Notably, LGDA is designed as a plug-and-play module that can be seamlessly integrated into existing recommendation frameworks without modifying the original system configurations. Extensive experiments on four real-world datasets demonstrate that LGDA effectively alleviates data sparsity and class imbalance, leading to significant improvements in both recommendation performance and model robustness. The main contributions of this paper are:

- This paper proposes a novel approach of converting the user-item interaction bipartite graph into a line graph, to reformulate the problems of data sparsity and class imbalance as labeled node scarcity and label distribution skew at the line graph level, respectively, opening up a new technical path for the unified solution to these two core challenges in recommender systems.
- Based on the above idea, we propose the Line Graph-based Data Augmentation (LGDA) strategy, a plug-and-play module that can be directly integrated into existing recommendation architectures to address data sparsity and class imbalance without adjusting the original configuration.
- Furthermore, our proposed targeted augmentation and confidence filtering mechanism ensures the generation of high-quality,

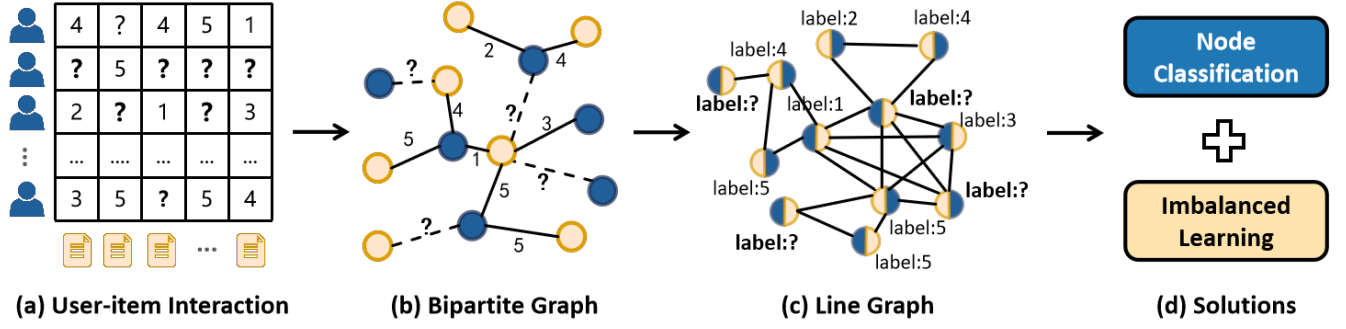


Figure 2: An intuitive illustration of our research motivation. Through line graph transformation, data sparsity in recommender systems is reformulated as the problem of numerous unlabeled nodes (marked "?"), while class imbalance is manifested as a skewed distribution of node labels (e.g., labels 4 and 5 dominate). This reformulation enables the direct application of node classification and imbalanced graph learning methods to address these respective challenges.

distributionally balanced enhanced data, further strengthening LGDA's capability to manage data sparsity and class imbalance.

2 Preliminary

This section describes two relevant tasks used in this paper.

2.1 Rating-based Recommendation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the set of users and items, respectively, with their interactions represented by a rating matrix $\mathcal{R} \in \mathbb{R}^{|\mathcal{U}| \times |\mathcal{I}|}$, where each entry $r_{u,i}$ corresponds to the observed rating of user $u \in \mathcal{U}$ on item $i \in \mathcal{I}$, while unobserved entries are denoted as $r_{u,i} = \emptyset$. Furthermore, the initial attributes of users and items are denoted by the matrices $X_{\mathcal{U}} \in \mathbb{R}^{|\mathcal{U}| \times d_u}$ and $X_{\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}| \times d_i}$, respectively, where d_u and d_i denote the dimensions of the user and item attributes. The goal of rating-based recommendation is to learn a function $\mathcal{F}$ that accurately predicts a user's rating $\hat{r}_{u,i}$ for an item based on their respective latent representations $\hat{e}_u$ and $\hat{e}_i$:

$$\hat{r}_{ui} = \mathcal{F}(\hat{e}_u, \hat{e}_i). \quad (1)$$

A common optimization objective for learning $\mathcal{F}$ is the Mean Squared Error (MSE) between predicted and true ratings:

$$\mathcal{L}_{MSE} = \frac{1}{|\mathcal{Z}|} \sum_{(u,i) \in \mathcal{Z}} (r_{u,i} - \hat{r}_{u,i})^2, \quad (2)$$

where $\mathcal{Z}$ is the set of observed user-item pairs in the training set, and $r_{u,i}$ is the ground-truth rating. However, against this backdrop, a key challenge stems from the inherent sparsity of real-world rating data. Specifically, the sparsity of the rating matrix is formally defined as $1 - \frac{|\mathcal{Z}|}{|\mathcal{U}| \times |\mathcal{I}|}$. When sparsity is high, the function $\mathcal{F}$ will struggle to learn meaningful representations, which will lead to degraded recommendation performance.

2.2 Node Classification

Formally, let graph $G = (V, E, X)$ be a graph with node set V , edge set E , and feature matrix X . Given a label set $Y = \{y_1, \dots, y_N\}$ where each $y_i \in \{C_1, \dots, C_N\}$, the goal of node classification is to learn a classifier f_θ that generalizes from the subset of labeled

nodes $Y_L \subset Y$ to predict the labels of the unlabeled nodes Y_U . A standard method employs message-passing to aggregate neighborhood information, with predictions made via:

$$\hat{y}_v = \text{softmax}(W_{\text{cls}} \cdot \sigma(W^{(l)} \cdot \text{AGG}^{(l)}(h_v^{k-1}, \forall v' \in \mathcal{N}(v)))), \quad (3)$$

where $\text{AGG}(\cdot)$ aggregates feature from the k -hop neighborhood $\mathcal{N}(v)$, W denotes learnable parameters, and σ is an activation function. Training typically minimizes the cross-entropy loss:

$$\mathcal{L}_{\text{node}} = - \sum_{v \in V_L} \sum_{c=1}^C y_{v,c} \cdot \log(\hat{y}_{v,c}), \quad (4)$$

where $y_{v,c}$ indicates the ground-truth label of node v . Nevertheless, in practice, label distributions are often highly skewed. Defining the class imbalance ratio as $\rho = \frac{\min_i |C_i|}{\max_i |C_i|}$, high values of ρ correspond to severe imbalance. In such cases, standard classifiers f_θ become biased toward majority classes, leading to degraded performance on minority-class nodes.

3 The Proposed Framework

This section details the proposed LGDA strategy, with its overall framework illustrated in Figure 3. Specifically, LGDA accepts the user attributes $X_{\mathcal{U}}$, item attributes $X_{\mathcal{I}}$, and rating matrix $\mathcal{R}$ as input.

3.1 Line Graph Transformation

Specifically, the line graph is constructed by the following process.

3.1.1 Augmented Target Selection. In this study, the original rating matrix $\mathcal{R}$ contains a large number of unobserved ratings. Directly treating all these unobserved entries as candidates for data augmentation and transforming them into a bipartite graph would result in a combinatorial explosion of nodes and edges. To address this, we propose a targeted augmentation mechanism. We focus on the long-tail groups—inactive users and unpopular items, defined as those with interaction frequencies below the global average—and selectively enhance only two specific types of potential interactions: "inactive user-popular item" and "unpopular item-active user". For each inactive user, we randomly generate links to $\gamma \times \text{avg}$ popular items, where avg denotes the global average interaction count. Likewise, for each unpopular item, we connect it to $\gamma \times \text{avg}$

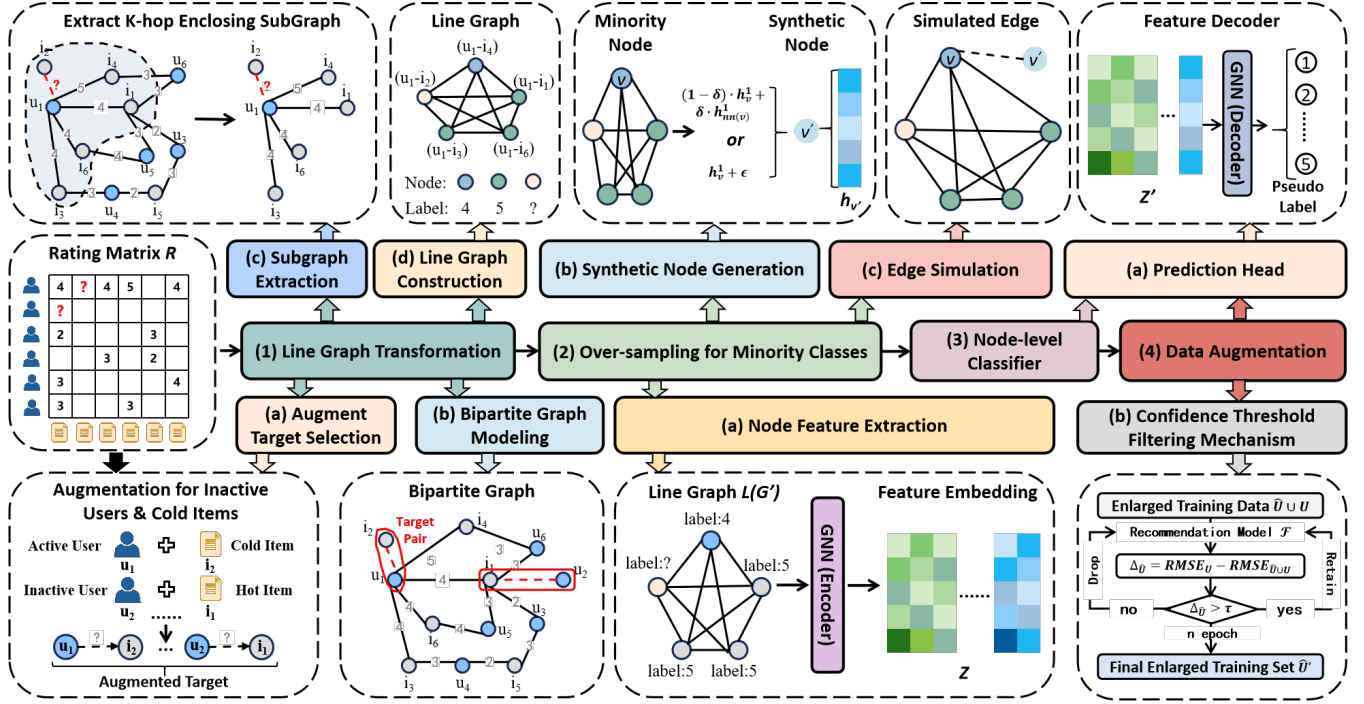


Figure 3: The main framework of the proposed LGDA strategy.

active users. The augmentation ratio γ is tuned to bring the average interaction frequency of these long-tail entities close to the global average, thereby mitigating data sparsity and imbalance.

3.1.2 Interaction Bipartite Graph Modeling. The augmented rating matrix $\hat{R}$, together with the user and item attribute matrices, is represented as a weighted attributed bipartite graph $G = (V, E, X, \mathcal{W})$. Here, the node set $V = \mathcal{U} \cup \mathcal{I}$ comprises all user and item nodes. Each edge $e = (u, i) \in E$ corresponds to an observed or augmented interaction, and its weight $\mathcal{W}(e)$ is directly taken from the corresponding entry in $\hat{R}$. The node feature matrix $X \in \mathbb{R}^{(|\mathcal{U}|+|\mathcal{I}|) \times (d_u+d_i)}$ is constructed by concatenating the user attribute matrix $X_u \in \mathbb{R}^{|\mathcal{U}| \times d_u}$ and the item attribute matrix $X_i \in \mathbb{R}^{|\mathcal{I}| \times d_i}$ along the feature dimension.

3.1.3 Enclosing Subgraph Extraction. We avoid directly converting the entire bipartite graph into a line graph, as this operation would lead to exponentially increasing computational complexity. To balance efficiency and performance, studies in [1] have shown that the existence of potential links can be effectively inferred from the local topological structure around target node pairs. Accordingly, for each target node pair (v_i, v_j) considered for augmentation, we extract its k -hop enclosing subgraph $G' = (V', E', \mathcal{W}', X')$ from the original bipartite graph G , where $V' = \{v \in V \mid \min(d(v, v_i), d(v, v_j)) \leq k\}$ is the node set, $E' = E \cap (V' \times V')$ is the edge set, $\mathcal{W}' = \{w_e \mid e \in E'\}$ is the edge weight set, and $X' = [x_v^T \mid v \in V'] \in \mathbb{R}^{|V'| \times d}$ is the feature matrix, with $d(\cdot)$ being the shortest path distance between two nodes, e an edge carrying weight w_e , and x_v the d -dimensional feature vector of node v .

3.1.4 Line Graph Transformation. Given the enclosing subgraph G' , its associated line graph $L(G')$ is constructed in three steps: (1) mapping each edge $e = (v_i, v_j)$ in G' to a node v_e in $L(G')$, with features concatenated from v_i and v_j ; (2) assigning the original edge weight w_e as the label of v_e ; and (3) connecting two nodes v_{e_i} and v_{e_j} in $L(G')$ if the corresponding edges e_i and e_j in G' share a common node. By converting the interaction bipartite graph G' into a line graph $L(G')$, the original data sparsity problem and class imbalance problem can be reformulated as the scarcity of labeled nodes and the skewness of label distribution, respectively.

3.2 Over-sampling for Minority Classes

Given the label imbalance in $L(G')$, we employ an oversampling module adapted from [27] to rebalance the class distribution.

3.2.1 Node Feature Extraction. To better capture both inter- and intra-class relationships for the synthetic minority-class nodes, we employ a single-layer GNN as the feature extractor. For each node v , its representation is computed as:

$$h_v^{(1)} = \sigma \left(W^{(1)} \cdot \text{AGG} (x_{v'} \mid v' \in \mathcal{N}(v) \cup v) \right), \quad (5)$$

where $x_{v'}$ is the feature of node v' , $\mathcal{N}(v)$ denotes the set of its neighbors, $\text{AGG}(\cdot)$ is a neighborhood aggregation function, $W^{(1)}$ is a learnable weight matrix, and σ is a non-linear activation function.

3.2.2 Synthetic Node Generation. To perform interpolation between a minority sample and its same-class neighbors in the embedding space, the first step is to find the nearest labeled node

that belongs to the same class as h_v^1 , which is defined as

$$nn(v) = \underset{u}{\operatorname{argmin}} \|h_u^1 - h_v^1\|, \quad (6)$$

where $nn(v)$ refers to the nearest neighbor of v from the same class. Note that $Y_u = Y_v$. However, subgraph partitioning can result in a limited number of node categories in the line graph, which may prevent the finding of same-class neighbors for minority-class nodes. To address this bottleneck, we generate synthetic nodes based on neighborhood availability. When a minority-class node v has same-class neighbors, a new node is generated by linearly interpolating its features with those of its neighbors. When no same-class neighbors are available, new samples are generated through intra-class interpolation by introducing Gaussian noise to existing minority-class samples. This process can be described as

$$h_{v'}^1 = \begin{cases} (1 - \delta) \cdot h_v^1 + \delta \cdot h_{nn(v)}^1 & \text{if } |nn(v)| > 0, \\ h_v^1 + \epsilon & \text{otherwise,} \end{cases} \quad (7)$$

where $\delta \in [0, 1]$ is a mixing coefficient that governs the weighting between the original node's features and those of its neighbor, and $\epsilon \sim N(0, \rho^2 I)$ denotes the Gaussian noise vector with mean 0 and variance $\rho^2 I$. Additionally, we introduce a hyper-parameter λ (the over-sampling scale) to control the number of synthetic samples generated for each minority class. Through this generation process, we can make the distribution of class size more balanced. Finally, we obtain the synthetic nodes V'' for the minority-class.

3.2.3 Edge Simulation. Synthetic nodes have been generated to balance the class distribution. However, these nodes remain isolated from the original graph due to the lack of connecting edges. To integrate them, we introduce an edge generator to simulate relational links. Specifically, each synthetic node is randomly connected to t existing nodes from the original graph:

$$E'' = E_L \cup \bigcup_{v' \in V''} \{(v', u) \mid u \sim \operatorname{Uniform}(V_L, t)\}, \quad (8)$$

where $\operatorname{Uniform}(\cdot)$ denotes a function that uniformly samples t nodes from the set V_L . In this work, we set $t = 3$. The newly generated nodes and edges are then incorporated into the adjacency matrix of the augmented line graph $\hat{L}(\hat{G}')$, which now possesses a balanced label distribution, and serve as input to the subsequent GNN-based classifier.

3.3 Node-level Classifier

Here, the balanced line graph $L(\hat{G}')$ will be used to train an unbiased GNN classifier f_θ . Notably, the backbone classifier f_θ of our proposed strategy can be implemented using any type of GNN. Once trained, f_θ can generate distribution-balanced pseudo-labels $\hat{Y}$ for unlabeled nodes. As previously mentioned, node labels in the line graph correspond directly to edges and their associated ratings in the original user-item bipartite graph. Hence, predicting pseudo-labels Y_U for unlabeled line graph nodes is equivalent to inferring potential user-item rating interactions. When these potential interactions are treated as augmented data, they not only alleviate the problem of training data sparsity but also address the issue of imbalanced rating distribution.

3.4 Data Augmentation and Refinement

The selection of augmented data should not be made arbitrarily. During the data augmentation process, if all data is introduced without careful screening, a large amount of noise is likely to be introduced. To ensure the quality of data augmentation, we design a confidence scoring mechanism that calculates a confidence score for each batch of augmented data. Let $\hat{U}$ be a batch of augmented data, the confidence is measured by:

$$\Delta_{\hat{U}} = RMSE_U - RMSE_{U \cup \hat{U}}, \quad (9)$$

where $\Delta_{\hat{U}}$ denotes the RMSE difference between the original training set U and the augmented training set $U \cup \hat{U}$. If $\Delta_{\hat{U}} > \tau$ (typically $\tau = 0$), the batch is considered confident and added to the training set; otherwise, it is discarded.

Table 1: The basic statistics of the four datasets.

Dataset	CiaoDVD	MovieLens 100k	SCHOLAT	Yelp
Users	17,616	943	32,182	43,874
Items	16,122	1,682	19,645	11,536
Ratings	72,665	100,000	680,267	21,5879
Sparsity	99.97%	93.69%	99.89%	99.95%
Imbalance ratio	12.33	5.59	22.59	4.57

4 Experiments

In this section, we systematically evaluate the proposed LGDA strategy on four real-world datasets to verify its effectiveness in mitigating data sparsity and class imbalance.

4.1 Experimental Settings

4.1.1 Datasets. Four widely used recommendation datasets are adopted for experiments: CiaoDVD¹, MovieLens 100k², SCHOLAT³, Yelp⁴. These datasets vary in terms of field, scale, sparsity, and imbalance ratios. The basic statistics and real rating distributions of the datasets are provided in Table 1 and Figure 7, respectively. See the Appendix for further dataset details.

4.1.2 Baselines. To evaluate the effectiveness of our LGDA strategy, we specifically integrate it into the following eight representative models: FM [18], DNN [5], xDeepFM [16], LightGCN [11], NCL [17], CGCL [10], SCCF [22] and RecDCL [23]. See the Appendix for further baseline details.

4.1.3 Evaluation Metrics. We evaluate performance using two standard metrics: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Lower values for both indicate better performance, and even marginal improvements can significantly enhance recommendation quality. See the Appendix for metric calculations.

4.2 Overall Comparison

In this section, we focus on the overall performance of various baseline algorithms with (w/) and without (w/o) LGDA augmentation across four real-world datasets. Table 2 reports the optimal results

¹<https://guoguibing.github.io/librec/datasets/CiaoDVD.zip>

²[https://files.grouplens.org/datasets/movielens/MovieLens 100k/](https://files.grouplens.org/datasets/movielens/MovieLens%20100k/)

³https://www.scholal.com/datasetApplication.html?dataset=user_recommendation

⁴<https://tianchi.aliyun.com/dataset/89459>

Table 2: Experimental results of the baseline algorithms with (w/) or without (w/o) our LGDA enhancement on the four datasets. The best results are boldfaced, with "Δ" denoting a reduction in MAE or RMSE.

Datasets	Metric	FM		DNN		xDeepFM		LightGCN		NCL		CGCL		SCCF		RecDCL	
		MAE	RSME	MAE	RSME	MAE	RSME	MAE	RSME	MAE	RSME	MAE	RSME	MAE	RSME	MAE	RSME
CiaoDVD	w/o	3.77	3.93	3.63	3.79	0.83	1.08	4.05	4.19	3.46	3.67	2.55	2.83	3.75	3.91	2.85	3.11
	w/	1.32	1.73	1.87	2.02	0.65	0.89	2.52	2.84	1.58	2.01	0.95	1.31	1.84	2.24	1.41	1.75
	Δ	2.45	2.20	1.76	1.77	0.18	0.19	1.53	1.35	1.88	1.66	1.60	1.52	1.91	1.67	1.44	1.36
MovieLens 100k	w/o	0.78	0.97	1.77	2.01	0.92	1.11	2.26	2.51	1.16	1.40	0.90	1.14	1.19	1.40	0.87	1.09
	w/	0.41	0.66	0.55	0.82	0.53	0.76	0.63	0.93	0.69	0.92	0.47	0.72	0.45	0.69	0.53	0.76
	Δ	0.37	0.31	1.22	1.19	0.39	0.35	1.63	1.58	0.47	0.48	0.43	0.42	0.74	0.71	0.34	0.33
SCHOLAT	w/o	0.57	0.85	0.49	0.73	0.51	0.73	1.12	1.40	0.69	0.97	0.52	0.78	0.65	0.86	0.48	0.74
	w/	0.34	0.60	0.39	0.66	0.36	0.61	0.52	0.84	0.51	0.77	0.47	0.72	0.37	0.61	0.43	0.68
	Δ	0.23	0.25	0.10	0.07	0.15	0.12	0.60	0.56	0.18	0.20	0.05	0.06	0.28	0.25	0.05	0.06
Yelp	w/o	0.98	1.25	1.20	1.52	0.89	1.13	2.13	2.58	1.29	1.61	1.07	1.34	1.15	1.45	1.03	1.30
	w/	0.48	0.83	0.62	0.92	0.49	0.81	0.92	1.46	0.66	1.04	0.65	0.96	0.53	0.84	0.58	0.87
	Δ	0.50	0.42	0.58	0.60	0.40	0.32	1.21	1.12	0.63	0.57	0.42	0.38	0.62	0.61	0.45	0.43

of the comparison experiment. Overall, LGDA significantly reduces MAE and RMSE across all datasets and baseline algorithms, validating its effectiveness as a general plug-and-play augment component to alleviate data sparsity and class imbalance and improve recommendation performance. In terms of data sparsity, LGDA achieves the most significant improvement on the sparsest dataset, CiaoDVD, while the gain is relatively smaller on the densest dataset, MovieLens 100k. This contrast indicates that LGDA's enhancement is more pronounced when data is sparser. Regarding class imbalance, the performance gain of LGDA follows a rise-then-decline trend across datasets with varying imbalance ratios. On low-imbalance datasets such as Yelp and MovieLens 100k, improvements are primarily due to the alleviation of data sparsity. In contrast, on moderately imbalanced datasets like CiaoDVD, LGDA performs best as its targeted augmentation effectively corrects prediction bias toward long-tail classes. Even in cases of high imbalance (e.g., SCHOLAT), where diminishing returns are observed, LGDA maintains stable gains through local oversampling, demonstrating its robustness.

4.3 Ablation Study

In this section, three independent studies are carried out to explore the contribution of each design component in our LGDA strategy.

4.3.1 Effect of Backbone Classifier. In this section, we select three classical graph neural network models (GCN [13], GraphSAGE [9], and GAT [20]) as backbone classifier to investigate the influence mechanisms of different neural network architectures on the data augmentation process. We focus on analyzing the correlation between the backbone classifier's pseudo-label accuracy (ACC), RMSE reduction (Δ RMSE), and the number of final augmented samples (Δ Samples). The results are presented in Table 4. The results show that Δ RMSE exhibits a positive correlation with ACC. Employing GraphSAGE as the backbone classifier yields higher Δ RMSE values in most scenarios, indicating superior quality of its generated pseudo labels and a more pronounced enhancement in recommendation performance. Notably, even when GAT achieves high pseudo label accuracy in certain cases, its Δ RMSE is not always optimal. This demonstrates that high ACC alone is insufficient to ensure the effective augmentation, and the reliability of augmented samples also needs to be considered. However, there is no straightforward

linear relationship between Δ RMSE and Δ RMSE. In some cases, fewer augmented samples even lead to a greater Δ RMSE. For example, GCN+RecDCL achieves Δ RMSE = 1.36 with only 45,056 samples on the CiaoDVD dataset. This indicates that sample quality is more important than quantity. Nevertheless, in GraphSAGE, higher Δ RMSE are often accompanied by higher Δ RMSE. This shows that on the premise of ensuring quality, increasing the number of samples is still beneficial for improving performance. Furthermore, the enhancement effects of different backbone classifiers on baseline models show significant differences. GCN demonstrates stable overall performance, particularly on the CiaoDVD and SCHOLAT datasets, where it achieves a certain degree of improvement for baseline models such as FM and LightGCN. GraphSAGE performs optimally on most baseline models and datasets, exhibiting clear advantages especially in terms of Δ RMSE and ACC. GAT shows relatively unstable performance. This may be because its attention mechanism is sensitive to noise.

4.3.2 Effect of Over-sampling Module. To evaluate the effectiveness of the over-sampling module, this section primarily examines how its inclusion (w/) or exclusion (w/o) impacts the RMSE performance of baseline models across all rating levels. The comparative results on four datasets are shown in Table 5. For the tail categories, the LGDA equipped with an over-sampling module leads to a notable decrease in RMSE for most models. Even for extremely rare classes (e.g., rating=-1 in SCHOLAT), all models show consistent "w/" > "w/o" improvements. This validates the stability of the LGDA strategy in tail-class optimization. In the head categories, most models achieve a reduction in RMSE. Although this decrease is smaller than that observed in tail categories, it shows that, while balancing data distribution, LGDA does not affect the stability of predictions for head categories. The "all" row across all datasets shows that the LGDA equipped with an over-sampling module significantly reduces the overall RMSE of all baseline models. Notably, even in the most imbalanced SCHOLAT dataset, all models benefit significantly by LGDA over-sampling, confirming its effectiveness under extreme data distributions. Furthermore, a small increase in RMSE is observed for a few models on specific classes (e.g., the FM model for rating=4 in MovieLens 100k), presumably due to slight

Table 3: Performance comparison of baselines when enhanced with (w/) versus without (w/o) the LGDA confidence threshold mechanism, across four datasets. The best results are boldfaced. $\Delta\text{RMSE} = \text{RMSE}_U - \text{RMSE}_{U \cup \hat{U}}$ denotes the reduction in RMSE. $\Delta\text{Samples}$ is the number of augmented samples.

Datasets	Metric	FM		DNN		xDeepFM		LightGCN		NCL		CGCL		SCCF		RecDCL	
		w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
CiaoDVD	ΔRMSE	2.20	2.20	1.01	1.77	-0.51	0.19	1.35	1.35	1.13	1.66	1.24	1.52	0.78	1.67	1.52	1.36
	$\Delta\text{Samples}$	112,108	112,108	112,108	12,288	112,108	12,288	112,108	112,108	112,108	86,016	112,108	95,724	112,108	108,012	112,108	45,056
MovieLens 100k	ΔRMSE	0.23	0.31	0.76	1.19	-0.01	0.35	1.42	1.58	0.15	0.48	0.36	0.42	0.65	0.71	-0.13	0.33
	$\Delta\text{Samples}$	131,638	131,072	131,638	86,016	131,638	69,632	131,638	131,638	131,638	114,688	131,638	118,784	131,638	122,880	131,638	82,486
SCHOLAT	ΔRMSE	0.12	0.25	0	0.07	-0.12	0.12	0.47	0.56	0.12	0.20	0.02	0.06	0.18	0.25	0	0.06
	$\Delta\text{Samples}$	1,253,356	380,928	1,253,356	65,536	1,253,356	98,304	1,253,356	684,032	1,253,356	262,144	1,253,356	53,248	1,253,356	376,832	1,253,356	86,016
Yelp	ΔRMSE	0.18	0.42	0.42	0.60	0.08	0.32	1.03	1.12	0.42	0.57	0.28	0.38	0.33	0.61	0.23	0.43
	$\Delta\text{Samples}$	301,754	248,506	301,754	150,202	301,754	196,608	301,754	293,562	301,754	233,472	301,754	135,168	301,754	281,274	301,754	195,258

pseudo-label distribution bias introduced by oversampling. However, this impact does not alter the overall trend of performance improvement. Therefore, we have sufficient reason to believe that the LGDA oversampling module can effectively alleviate class imbalance by specifically supplementing tail samples and balancing data distribution.

4.3.3 Effect of Confidence Threshold Mechanism. To verify the effectiveness of the pseudo-label confidence threshold mechanism, this section investigates the impact of enabling (w/) versus disabling (w/o) the mechanism on both the RMSE reduction (ΔRMSE) and the number of final augmented samples ($\Delta\text{Samples}$) of baseline models. The comparative results on four datasets are presented in Table 3. Obviously, there are significant differences in ΔRMSE and $\Delta\text{Samples}$ between enabling (w/) and disabling (w/o) the confidence threshold mechanism. When the mechanism is disabled, all $\Delta\text{Samples}$ are retained, with ΔRMSE remaining at a relatively low level. In contrast, when enabling the mechanism, the number of $\Delta\text{Samples}$ decreases sharply, and the ΔRMSE rises significantly. This phenomenon aligns with our expectations. Because when the mechanism is disabled, it is highly likely that a large number of low-confidence noisy samples are mixed into the augmented samples. These samples can seriously interfere with the model's prediction performance and even cause some models to exhibit negative ΔRMSE values. On the contrary, after enabling the threshold mechanism, it can filter out low-confidence pseudo-labels (i.e., noisy samples) while retaining "high-confidence, low-noise" effective samples. As a result, the phenomenon of negative ΔRMSE is eliminated. However, a few models show a slight decrease in ΔRMSE after enabling the threshold (for example, the RecDCL model in the CiaoDVD dataset, with its ΔRMSE dropping from 1.52 to 1.36). It is speculated that the reason may be an inappropriate threshold setting, which filters out some useful samples and reduces the effective information that the model can learn. But this limitation can be avoided by setting the threshold reasonably. Overall, a substantial amount of solid evidence indicates that this mechanism can effectively address the core issue of noise introduction in traditional data augmentation, providing reliable support for robust data augmentation in recommendation systems.

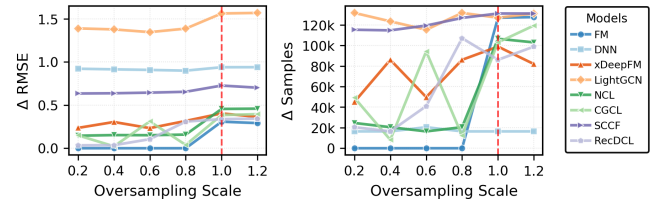


Figure 4: The enhancement effect of the LGDA strategy under varying over-sampling scale λ .

4.4 Hyper-Parameter Sensitivity Analysis

This section evaluates the sensitivity of LGDA to its core hyper-parameters: the oversampling scale λ , the confidence threshold τ , and the augmentation ratio γ . All experiments are conducted in a controlled environment using the MovieLens 100k dataset with GAT as the backbone classifier. We report the RMSE reduction (ΔRMSE) alongside the number of generated samples ($\Delta\text{Samples}$) across different parameter configurations.

4.4.1 Over-sampling Scale λ . To investigate the enhancement effect of the LGDA strategy under different over-sampling scales λ , we fixed the confidence threshold $\tau = 0$ and the data augmentation ratio $\gamma = 1.0$, and conducted experiments with $\lambda \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2\}$. The experiment results are shown in Figure 4. It is apparent that when λ reaches 1.0, both ΔRMSE and $\Delta\text{Samples}$ reach their peaks, demonstrating that LGDA achieves the most remarkable improvement in recommendation performance under this setting. However, when λ further increases to 1.2, both metrics decline to varying degrees. This phenomenon reveals that excessive oversampling does not yield better performance. A moderate oversampling scale (e.g., $\lambda = 1.0$) can effectively alleviate data sparsity and enhance the model's generalization ability on long-tail items. In contrast, an excessive oversampling scale (i.e., $\lambda = 1.2$) may introduce redundant information and noisy samples, leading to model overfitting or learning bias, and thus degrading recommendation accuracy.

4.4.2 Confidence Threshold τ . To evaluate how the confidence thresholds τ affects LGDA's performance, we set the over-sampling scale $\lambda = 1.0$ and the data augmentation ratio $\gamma = 1.0$, and evaluated τ across $\{0, 0.02, 0.04, 0.06, 0.08, 0.1\}$. The experiment results

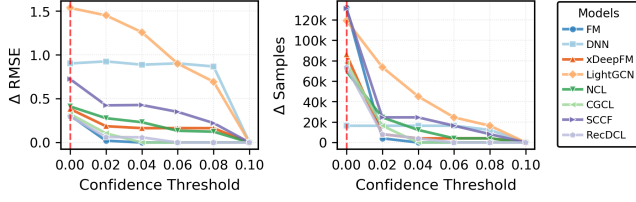


Figure 5: The enhancement effect of the LGDA strategy under varying confidence thresholds τ .

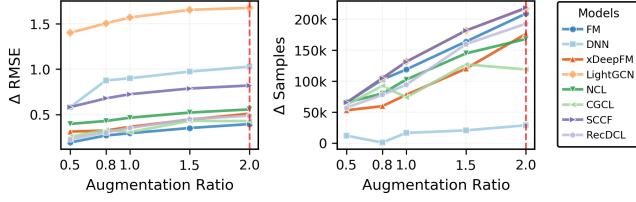


Figure 6: The enhancement effect of the LGDA strategy under varying augmentation ratio γ .

are shown in Figure 5. Evidently, variations in the confidence threshold τ exert a significant influence on both ΔRMSE and $\Delta\text{Samples}$. It is worth noting that although a lower confidence threshold, while potentially increasing the sample usage, may not necessarily lead to a higher ΔRMSE . Conversely, moderately raising the confidence threshold, although reducing the sample usage, may stabilize the effect of model performance improvement. Therefore, when it comes to practical applications, it is crucial to choose the right confidence threshold, balancing sample usage with the need to enhance model performance.

4.4.3 Augmentation Ratio γ . To evaluate how the data augmentation ratio γ affects LGDA's performance, we set the over-sampling scale $\lambda = 1.0$ and the confidence threshold $\tau = 0$, and evaluated γ across $\{0.5, 0.8, 1.0, 1.5, 2.0\}$. The experiment results are shown in Figure 6. The overall trend shows that as γ increases from 0.5 to 1.5, both ΔRMSE and $\Delta\text{Samples}$ rise significantly, indicating that the augmented data effectively enriches long-tail patterns without introducing substantial noise. However, when γ reaches 2.0, some models exhibit a decline in both metrics, suggesting that excessive augmentation may generate redundant or spuriously correlated samples, which impairs generalization. These results demonstrate that while raising γ alleviates data sparsity, the benefits exhibit a saturation point. Hence, in practice, the optimal γ should be selected by balancing data utility and model performance, rather than blindly pursuing a higher augmentation ratio.

5 Related Work

5.1 Line Graph-based Methods

With the advancement of graph neural networks, the structural properties of line graphs have been increasingly applied to various tasks. LGLP [1] converts original graphs to line graphs to model edge-edge interactions explicitly, facilitating target link feature

learning. LGCL [25] reformulates link prediction as node classification through contrastive learning and proposes a subgraph-to-line-graph node contrastive paradigm to reduce information redundancy. DLGNN [7] directly generates edge embeddings via line graph structures to enhance information aggregation in graph convolution. LGSMOTE-IDS [26] resolves class imbalance and the neglect of edge features by converting the fine-grained protocol service graph in network intrusion detection into a corresponding line graph. However, current research primarily focuses on node-level tasks, and integrating line graphs with GNNs to address data sparsity remains an under-explored area.

5.2 Class-Imbalanced Graph Learning

With the widespread application of GNNs for modeling complex graph structures, the class imbalance problem has become increasingly prominent in graph data mining. Existing approaches address this issue from diverse perspectives. Several methods, such as GraphSMOTE [27] and FincGAN [12], focus on data-level augmentation by generating synthetic minority-class samples—either via interpolation-based node synthesis or generative adversarial networks in heterogeneous graphs. Another line of work, including ReNode [2] and IceBerg [15], operates at the algorithmic level. ReNode introduces a topological imbalance metric to adjust node weights adaptively, while IceBerg employs a double-balancing mechanism to counteract the Matthew effect and label shift in self-training. Other methods like DR-GCN [19] and G2GNN [21] incorporate regularization and contrastive learning strategies to enhance model generalization under imbalance. Additionally, GraphSR [30] enhances minority-class diversity by selectively sampling informative unlabeled nodes.

6 Conclusion

Adopting a line graph perspective, this research provides novel insights into data sparsity and class imbalance in recommender systems. By converting the user-item interaction bipartite graph into a line graph, these two originally challenging problems are transformed into "insufficient labeled nodes" and "unevenly distributed labels" in the line graph, respectively. Building on this reformulation, this paper further proposes a Line Graph-based Data Augmentation (LGDA) strategy to address data sparsity and imbalance issues in a plug-and-play manner. Extensive experiments on four real-world datasets have verified the effectiveness and applicability of LGDA in mitigating data sparsity and class imbalance issues. While LGDA shows considerable promise, it also presents exciting opportunities for future development. For instance, future work could explore incorporating temporal dynamics and sequential patterns to capture evolving user preferences. Additionally, investigating scenarios with unknown or shifting data distributions represents another promising direction to enhance the model's robustness and generalizability.

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A Appendix

A.1 Full Pipeline Procedure

The full pipeline of LGDA can be summarized in Algorithm 1.

Algorithm 1 The full pipeline of LGDA

Input: User attributes X_U , Item attributes X_I , Rating matrix $\mathcal{R}$, Oversampling scale λ , Augmentation ratio γ , Confidence threshold τ

Output: Augmented training set $\mathcal{D}_{aug}$

- 1: **Phase 1: Line Graph Construction**
- 2: Let $avg \leftarrow$ global average interaction count;
- 3: **for** inactive users $u \in \mathcal{U}$ and unpopular items $i \in \mathcal{I}$ **do**
- 4: Select two target types of pairs from $\mathcal{R}$ for augmentation
- 5: (inactive-user-popular-item and cold-item-active-user);
- 6: Enhanced $\mathcal{R}$ until $|\mathcal{R}_u|$ and $|\mathcal{R}_i|$ equals to $avg \times \gamma$;
- 7: **end for**
- 8: Build attributed weighted bipartite graph G ;
- 9: **for** each target node pair (v_i, v_j) **do**
- 10: Extract k -hop enclosing subgraph G' from G ;
- 11: Transform subgraph G' to line graph $L(G')$;
- 12: **end for**
- 13: **Phase 2: Over-sampling for Minority Classes**
- 14: **for** each line graph $L(G')$ **do**
- 15: **for** each minority class c where $|Y_L^c| < \lambda \cdot \max_c |Y_L^c|$ **do**
- 16: Generate synthetic nodes and their edges via Eq.(5)-(8);
- 17: **end for**
- 18: **end for**
- 19: **Phase 3: Classifier Training**
- 20: Train an unbiased node classifier f_θ on $L(\hat{G}')$ via Eq.(3)-(4);
- 21: **Phase 4: Data Augmentation & Refinement**
- 22: **for** each mini-batch $\hat{U}$ of $\mathcal{R}'$ **do**
- 23: Get the confidence score $\Delta_{\hat{U}}$ of $\hat{U}$ by Eq.(9);
- 24: **if** $\Delta_{\hat{U}} > \tau$ **then**
- 25: $\mathcal{D}_{aug} \leftarrow \mathcal{D}_{aug} \cup \hat{U}$;
- 26: **end if**
- 27: **end for**
- 28: **return** $\mathcal{D}_{aug}$

A.2 Time Complexity Analysis

The overall time complexity of the LGDA strategy is analyzed in a modular manner. The complexity of each core module is as follows:

A.2.1 Line Graph Construction Process. This process consists of three stages:

- **Interactive Bipartite Graph Modeling.** Its complexity is determined by the number of input interactions $|E|$, resulting in a time complexity of $O(|E|)$.
- **Enclosing Subgraph Extraction.** For each target node pair to be augmented (totaling T pairs), a k -hop enclosing subgraph is extracted to control the global computational cost. The complexity is proportional to T and the average size of the subgraph $|S_k|$, i.e., $O(T \cdot |S_k|)$.
- **Subgraph-to-Line-Graph Conversion.** Converting each subgraph into a line graph involves two core operations: mapping

subgraph edges (totaling $|E'|$) into line graph nodes, and checking adjacency relationships between line graph nodes. The worst-case complexity for adjacency verification is $O(|E'|^2)$. Extending to T subgraphs, the overall complexity of this stage is $O(T \cdot |E'|^2)$.

A.2.2 Oversampling for Minority Classes. This module generates synthetic minority-class nodes via GNN embedding interpolation. Its complexity includes two parts: embedding interpolation for M minority-class nodes, with a time complexity of $O(M)$, and constructing random edges for the generated $|V''|$ synthetic nodes, with a time complexity of $O(|V''|)$. Thus, the total complexity is $O(M + |V''|)$.

A.2.3 Training Node-Level Classifier. Training an L -layer GNN classifier relies on message passing and feature aggregation over the edges of the balanced line graph. Its complexity mainly depends on the number of GNN layers L and the total number of edges in the line graph $|E_L|$, yielding a time complexity of $O(L \cdot |E_L|)$.

A.2.4 Data Augmentation and Refinement. This step involves confidence based filtering of the augmented data, and its time complexity is determined by the scale of the augmented data $|D_{aug}|$, i.e., $O(|D_{aug}|)$.

A.2.5 Overall Complexity. In summary, the total time complexity of the LGDA strategy is the sum of all modules:

$$O(|E| + T \cdot |S_k| + T \cdot |E'|^2 + M + |V''| + L \cdot |E_L| + |D_{aug}|). \quad (10)$$

In practical deployment, the computational load separates into two phases: the offline preprocessing, dominated by line graph construction ($O(T \cdot |E'|^2)$), and the online training/fine-tuning, dominated by the GNN classifier training ($O(L \cdot |E_L|)$). The costs of oversampling, augmentation, and refinement are relatively low and often negligible in comparison. Therefore, the dominant terms of the time complexity can be simplified to:

$$O(T \cdot |E'|^2 + L \cdot |E_L|). \quad (11)$$

A.3 Dataset Description

Four widely used datasets are adopted for experiments:

- **CiaoDVD.** Derived from the now-defunct UK-based DVD review website Ciao. It contains ratings (1–5) from 17,616 users on 16,122 DVDs, along with textual reviews and explicit social trust relationships among users. The dataset exhibits both interaction sparsity and social network structure, and is commonly used to evaluate social recommendation and rating prediction.
- **MovieLens 100k.** Released by the GroupLens research group, it contains 100,000 explicit ratings (on a 1-5 scale) from 943 users on 1,682 movies. Besides interaction data, it provides user demographic information and movie metadata, serving as a classic benchmark for recommendation evaluation.
- **SCHOLAT.** Derived from the SCHOLAT academic social network, it consists of 680,267 historical interactions between 32,182 users and 19,645 items, with ratings in a ternary scheme of $\{-1, 1, 3\}$ indicating "disinterest", "interest", and "strong interest". The dataset supplies user and item attribute information, and its highly sparse and markedly long-tailed distribution

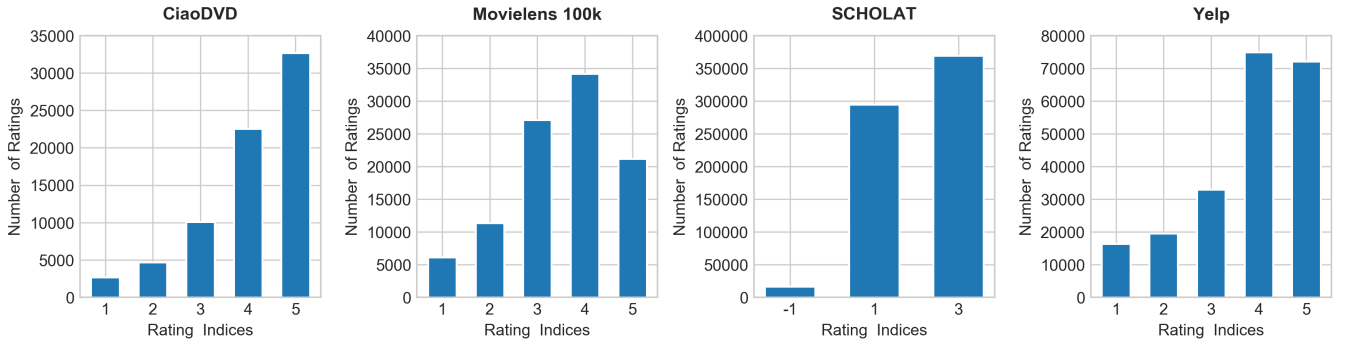


Figure 7: Real rating distribution across four datasets.

makes it an ideal benchmark for studying sparsity and imbalance in academic recommendation.

- **Yelp**. Sourced from the Yelp Open Dataset, it includes 215,879 ratings (1-5 stars) and reviews from 43,874 users on 11,536 businesses. It encompasses rich user profiles, business attributes, and geographical information, realistically simulating the challenges of data sparsity and skewed rating distributions in commercial local-service recommendation.

These datasets exhibit considerable diversity in domain, scale, sparsity, and class-imbalance ratio.

A.4 Baseline Introduction

To evaluate the effectiveness of our LGDA strategy, we specifically integrate it into the following eight representative models:

- **FM** [18]: A classic recommendation algorithm that efficiently captures second-order feature interactions via latent vector dot products, resolving linear models' non-linear combination limitations.
- **DNN** [5]: The model automatically mines complex nonlinear feature interactions in high-dimensional sparse data via a multi-layer network architecture, breaking the limitation of traditional shallow models in expressing feature relationships.
- **xDeepFM** [16]: This method employs an innovative Compressed Interaction Network (CIN) architecture to explicitly model high-order feature interactions, overcoming the limitation of traditional FM models which are only capable of capturing second-order interactions.
- **LightGCN** [11]: A classical GCN-based recommendation model that simplifies the original GCN design for CF tasks. It achieves iterative embedding propagation through user-item interaction graphs to precisely capture latent preference associations.
- **NCL** [17]: A novel paradigm of neighbor-enriched contrastive learning. It provides additional supervisory signals for recommendation tasks by constructing structural neighbors and semantic neighbors.
- **CGCL** [10]: This method represents a novel candidate-aware graph contrastive learning paradigm for recommendation, utilizing intermediate embeddings from different GCN layers as contrastive views and overcoming the limitation of information loss caused by traditional GCL's random data augmentation.

- **SCCF** [22]: This method employs a simple embedding model paired with an improved second-order cosine similarity contrastive loss, avoiding complex graph convolutional structures and offering a more concise and efficient paradigm for recommendation algorithm design.
- **RecDCL** [23]: An innovative dual contrastive learning recommendation framework integrates batch contrastive learning (BCL) and feature contrastive learning (FCL), overcoming the single-dimensional constraints of batch contrast in traditional self-supervised recommendation.

A.5 Evaluation Metrics

We use Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) to measure the accuracy of rating prediction, which are defined as

$$RMSE = \sqrt{\frac{\sum_{(u,i) \in \mathcal{T}} (r_{ui} - \hat{r}_{ui})^2}{|\mathcal{T}|}}, \quad (12)$$

$$MAE = \frac{\sum_{(u,i) \in \mathcal{T}} |r_{ui} - \hat{r}_{ui}|}{|\mathcal{T}|}, \quad (13)$$

where $\mathcal{T}$ denotes the testing set. r_{ui} and $\hat{r}_{ui}$ are the true and predicted ratings, respectively. The lower the RMSE and MAE values are, the better the prediction performance will be. Even a minor reduction in the RMSE and MAE values can have a considerable effect on the quality of the recommendations.

A.6 Implementation Details

We implement LGDA in Pytorch. The default optimizer is Adam with the learning rate of $1e-3$, and initialization is done via the Xavier method. For each preprocessed dataset, the ratio of training, validation, and test set is $6 : 2 : 2$. For the baselines, all methods are implemented using Recbole [29], with their hyperparameters following original paper settings or tuned via grid search. For our LGDA strategy, we set the enclosing subgraph hop k to 2, with the over-sampling scale λ adjusted within the range $\{0.2, 0.4, 0.6, 0.8, 1.0, 1.2\}$, the confidence threshold τ varied within the range $\{0, 0.02, 0.04, 0.06, 0.08, 0.1\}$, and the data augmentation ratio γ optimized within the range $\{0.5, 0.8, 1.0, 1.5, 2.0\}$. All experiments are conducted on a NVIDIA Tesla v100 GPU.

Table 4: Performance comparison of baselines with LGDA enhancement by using different backbone classifier. The best results are boldfaced. $\Delta\text{RMSE} = \text{RMSE}_U - \text{RMSE}_{U \cup \hat{U}}$ denotes the reduction in RMSE. $\Delta\text{Samples}$ is the number of augmented samples.

Backbone	Baseline	CiaoDVD			MovieLens 100k			SCHOLAT			YELP		
		ACC	ΔRMSE	$\Delta\text{Samples}$	ACC	ΔRMSE	$\Delta\text{Samples}$	ACC	ΔRMSE	$\Delta\text{Samples}$	ACC	ΔRMSE	$\Delta\text{Samples}$
GCN	FM		1.94	108,012		0	0		0.21	290,816		0.37	229,376
	DNN		1.61	12,288		0.88	16,384		0.06	45,056		0.59	162,490
	xDeepFM		0.02	4,096		0.12	8,192		0.12	98,304		0.18	102,400
	LightGCN	0.62	1.28	112,108	0.64	1.23	82,486	0.92	0.55	647,168	0.70	1.08	274,432
	NCL		1.51	86,016		0.16	20,480		0.17	208,896		0.57	233,472
	CGCL		1.21	81,920		0.01	4,096		0.04	24,576		0.16	53,248
	SCCF		1.53	108,012		0.44	106,496		0.18	221,184		0.53	240,314
	RecDCL		1.36	45,056		0.02	4,096		0.06	86,016		0.39	188,416
GraphSage	FM		2.2	112,108		0	0		0.23	352,256		0.40	217,088
	DNN		1.77	12,288		1.19	86,016		0.07	65,536		0.60	150,202
	xDeepFM		0.19	12,288		0.26	66,102		0.10	53,248		0.32	196,608
	LightGCN	0.97	1.35	112,108	0.98	1.39	131,072	0.99	0.52	569,344	0.95	1.12	293,562
	NCL		1.66	86,016		0.17	20,480		0.20	262,144		0.51	196,608
	CGCL		1.52	95,724		0.15	53,814		0.06	49,152		0.38	135,168
	SCCF		1.67	108,012		0.65	131,638		0.25	376,832		0.60	270,336
	RecDCL		1.23	45,056		0.04	12,288		0.05	53,248		0.40	170,682
GAT	FM		0.53	67,052		0.31	131,072		0.25	380,928		0.42	248,506
	DNN		2.71	12,288		0.91	16,384		0.06	40,960		0.56	131,072
	xDeepFM		0.17	12,288		0.35	69,632		0.11	77,824		0.28	166,586
	LightGCN	0.80	0	24,576	0.93	1.58	131,638	0.95	0.56	684,032	0.92	1.11	285,370
	NCL		0.16	24,576		0.48	114,688		0.17	212,992		0.56	221,184
	CGCL		0.71	40,960		0.42	118,784		0.06	53,248		0.34	117,434
	SCCF		0.27	58,860		0.71	122,880		0.23	331,776		0.61	281,274
	RecDCL		1.13	45,056		0.33	82,486		0.04	45,056		0.43	195,258

Table 5: The comparison of RMSE metric for baselines when enhanced with (w/) versus without (w/o) the LGDA over-sampling module, across four datasets. The best results are boldfaced. ‘tail’ and ‘head’ denote the tail class and head class respectively.

Datasets	Predicted Rating	FM		DNN		xDeepFM		LightGCN		NCL		CGCL		SCCF		RecDCL	
		w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
CiaoDVD	1 (tail)	0.70	0.43	0.54	0.23	3.09	2.66	0.98	0.92	0.61	0.52	1.58	1.19	0.70	0.26	1.03	1.70
	2	1.69	0.72	1.54	0.77	2.10	1.67	1.98	1.90	1.37	1.07	1.22	0.57	1.68	1.20	1.02	0.93
	3	2.69	1.30	2.53	1.76	1.10	0.68	2.98	2.76	2.28	1.61	1.55	1.03	2.67	1.97	1.68	1.78
	4	3.70	2.33	3.52	2.75	0.10	0.28	3.98	3.77	3.41	2.60	2.49	1.51	3.68	2.98	2.79	1.55
	5 (head)	4.72	3.37	4.52	3.74	0.90	1.23	4.99	4.79	4.48	3.67	3.54	2.28	4.70	3.99	3.89	2.12
	all	3.93	1.74	3.75	2.8	1.08	1.02	4.19	2.85	3.67	2.11	2.84	1.31	3.91	2.21	3.11	1.60
MovieLens 100k	1 (tail)	1.81	0.48	1.12	0.53	2.60	0.66	0.61	0.78	2.05	1.94	2.28	1.88	1.44	0.49	2.18	1.46
	2	1.14	1.21	1.03	1.18	1.62	1.17	0.90	0.83	1.04	0.59	1.50	0.43	0.61	1.23	1.42	1.19
	3	0.57	0.87	1.57	0.85	0.63	0.79	1.78	0.82	0.58	0.38	0.77	0.6	0.45	0.87	0.70	0.58
	4 (head)	0.66	0.82	2.13	0.85	0.36	0.80	2.71	1.43	1.23	1.01	0.70	0.75	1.35	0.82	0.65	0.73
	5	1.29	1.30	2.76	1.40	1.34	1.31	3.65	2.27	2.12	1.91	1.39	1.45	2.31	1.37	1.35	1.46
	all	0.96	0.75	2.00	0.83	1.11	0.85	2.51	1.11	1.39	0.92	1.14	0.73	1.40	0.76	1.08	1.06
SCHOLAT	-1 (tail)	2.07	2.05	2.10	2.05	2.53	2.18	1.85	2.01	2.14	2.10	2.24	2.20	2.06	2.04	2.24	2.16
	1	0.51	0.35	0.51	0.47	0.59	0.48	0.63	0.44	0.70	0.50	0.64	0.61	0.59	0.47	0.61	0.55
	3 (head)	0.98	0.83	0.74	0.74	0.67	0.61	1.79	1.41	1.07	0.99	0.77	0.75	0.96	0.82	0.72	0.71
	all	0.86	0.61	0.72	0.68	0.74	0.62	1.40	0.84	0.97	0.77	0.78	0.75	0.86	0.68	0.74	0.71
Yelp	1 (tail)	2.09	0.56	1.70	0.75	2.42	0.67	1.46	0.56	1.87	0.56	2.05	0.76	1.87	0.64	1.87	0.64
	2	1.48	1.50	1.10	1.59	1.60	1.58	1.66	1.44	1.56	1.50	1.55	1.54	1.50	1.47	1.50	1.47
	3	0.83	0.84	0.96	0.89	0.85	0.83	1.81	1.48	1.14	1.00	0.97	0.87	0.96	0.87	0.96	0.87
	4 (head)	0.76	0.75	1.32	0.68	0.63	0.63	2.30	1.90	1.17	1.01	0.89	0.71	0.93	0.90	0.93	0.90
	5	1.46	1.34	1.93	1.23	1.14	1.14	3.44	2.94	2.10	1.92	1.66	1.41	1.87	1.49	1.87	1.49
	all	1.24	0.83	1.52	0.92	1.16	0.83	2.58	1.46	1.62	1.06	1.36	0.98	1.44	0.82	1.44	0.82